DataLab

Grow Your Own Psychology Data

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Welcome to PS50008A

Research Methods & Experimental Design

Your first session of the module

What is DataLab?

Today you generate the dataset we analyse all term.

- You’ll be both **participant** and **tester**
- Work in pairs
- Follow your **sequence card**
- Complete 5 tasks + memory test

Session Overview

Time	Activity
10:00–10:25	Welcome, Part 1 survey, Memory Test
10:25–11:25	Complete 5 tasks (follow your sequence card)
11:25–11:35	Part 2 survey & wrap-up
11:35-11:55	Data Dictionary preview and discussion

What You'll Do

1. Complete Part 1 survey (demographics, mood, personality)
2. Do the **Memory Test** together
3. Follow your **sequence card** through 5 tasks
4. Complete Part 2 survey

Part 1: Getting Started

Get Your Materials

1. Collect your **lab notebook** (with your number)
2. Form into pairs (at least) max group size 4!
3. Collect a sequence card from your lab instructor

Complete Part 1 Survey

Scan the QR code below or open the link on the VLE:

[Qualtrics link displayed here]

Complete:

- Consent
- Sleep & lifestyle
- Current mood
- Personality (TIPI)
- Demographics

Memory Test

Mind Palace

We do this **together** as a group.

Two word lists on screen:

- **LEFT** = List A
- **RIGHT** = List B

Your instructor tells you which side.

The Procedure

1. **STUDY** (30 sec) — memorise YOUR list
2. **DISTRACT** (60 sec) — maths problems
3. **RECALL** (90 sec) — write on paper
4. **RECORD** — enter in Qualtrics

Ready?

Your instructor will say:

“Look LEFT” or **“Look RIGHT”**

Have pen and paper ready!

STUDY: Lab 304A (Mozart)

00:30

LEFT	RIGHT
WISDOM	HOPE
COURAGE	ANGER
JUSTICE	FAITH
FREEDOM	POWER
PATIENCE	GRACE
LOYALTY	VIRTUE
BEAUTY	DREAM
SPIRIT	PEACE
HONOUR	KNOWLEDGE
TRUTH	SOUL

STUDY: Lab 304B (Silence)

00:30

LEFT	RIGHT
APPLE	WINDOW
TABLE	RABBIT
CHAIR	JACKET
RIVER	CANDLE
MOUNTAIN	LADDER
GUITAR	BASKET
CLOCK	HAMMER
PENCIL	PILLOW
FLOWER	GARDEN
BOOK	BOTTLE

DISTRACT

01:00

Maths problems — no calculator!

$7 \times 8 = ?$ $15 + 27 = ?$ $64 \div 8 = ?$

$23 - 9 = ?$ $6 \times 9 = ?$ $81 \div 9 = ?$

$12 + 38 = ?$ $45 - 17 = ?$ $8 \times 7 = ?$

RECALL

01:30

Write your words on paper!

Don’t look at the screen.

Record Results

In Qualtrics, enter:

- Words recalled (0–10)
- Which positions you remembered
- Any intrusions (wrong words)
- Strategy used

Task Previews

The 5 Tasks

Follow your **sequence card**.

Each task: ~15 minutes

Work in pairs — swap roles halfway.

Task 1: Reflex Test

How fast is your nervous system?

- Partner drops ruler without warning
- Catch it as fast as you can
- Record the distance (cm)
- 5 trials each

Task 2: Time Warp

How accurate is your internal clock?

- Estimate when 60 seconds pass
- Estimate line lengths (visual)
- Estimate string lengths (touch only)

Task 3: Sixth Sense

Can you detect being watched?

- Sit back-to-back
- Partner flips coin: stare or look away
- Guess which, rate confidence
- 6 trials each

Task 4: Lady Luck

How random can you be?

- Say 20 “random” numbers (1–10)
- Roll die 10 times
- Flip coin 10 times

Task 5: Frontal Lob

How well do you know your ability?

- Rate your throwing (1–10)
- Throw 10 balls at basket
- Compare prediction to reality

How Tasks Work

1. Check your **sequence card** for task order
2. Go to that task area
3. Complete with your partner
4. **Enter data in Qualtrics**
5. Move to next task on your card

Begin Tasks!

Check Your Sequence Card

Find your first task.

Go there now with your partner.

Task Time: 15 minutes

15:00

Move to Next Task

Check your sequence card.

Enter data before moving!

Task Time: 15 minutes

15:00

Move to Next Task

Check your sequence card.

Enter data before moving!

Task Time: 15 minutes

15:00

Move to Next Task

Check your sequence card.

Enter data before moving!

Task Time: 15 minutes

15:00

Move to Next Task

Check your sequence card.

Enter data before moving!

Task Time: 15 minutes (Final)

15:00

Part 2: Wrap-up

Complete Part 2 Survey

Return to Qualtrics:

- Mood NOW (compare to before)
- Favourite task
- Least favourite task
- Comments

Pre-Post Design

We measured mood **before** and **after**.

This lets us see if DataLab affected how you feel.

Each person is their own control.

What Did We Learn?

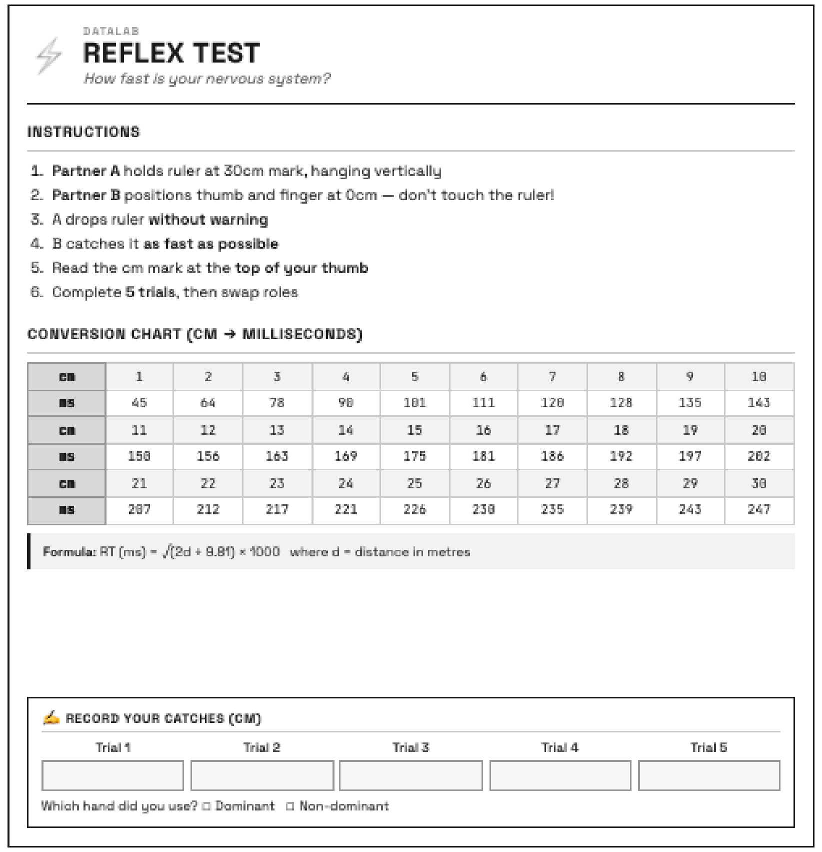
Reflex Test

Physics measures your brain!

- Distance fallen → reaction time
- Typical: 150–250 milliseconds
- Did you improve across trials?

What might have affected your data?

- Ruler height consistency?
- Anticipation of the drop?
- Partner's technique varied?



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REFLEX TEST

How fast is your nervous system?

INSTRUCTIONS

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
4. B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**
6. Complete **5 trials**, then swap roles

CONVERSION CHART (CM → MILLISECONDS)

cm	1	2	3	4	5	6	7	8	9	10
ms	45	64	78	90	101	111	120	128	135	143
cm	11	12	13	14	15	16	17	18	19	20
ms	150	156	163	169	175	181	186	192	197	202
cm	21	22	23	24	25	26	27	28	29	30
ms	207	212	217	221	226	230	235	239	243	247

Formula: RT (ms) = $\sqrt{(2d \div 9.81)} \times 1000$ where d = distance in metres

✍ **RECORD YOUR CATCHES (CM)**

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5

Which hand did you use? ☐ Dominant ☐ Non-dominant

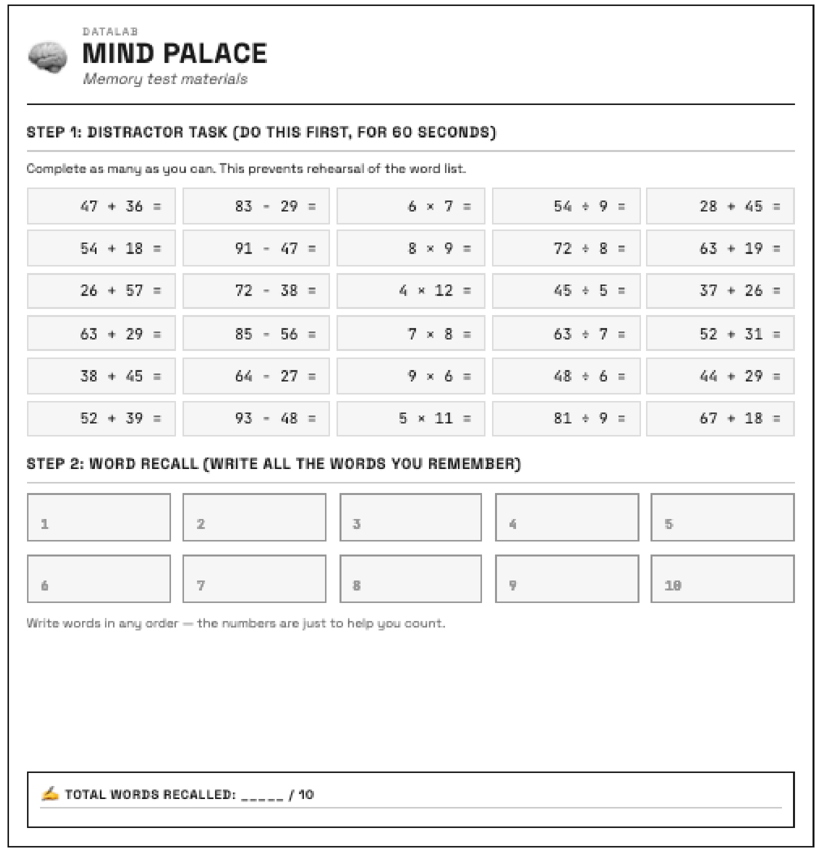
Mind Palace

Working memory in action

- Miller’s “magical number 7 ± 2”
- Serial position effect: primacy & recency
- Distraction task interfered with rehearsal

What might have affected your data?

- Could everyone hear clearly?
- Did you use a strategy (chunking)?
- Was the distraction task equally hard?



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MIND PALACE

Memory test materials

STEP 1: DISTRACTOR TASK (DO THIS FIRST, FOR 60 SECONDS)

Complete as many as you can. This prevents rehearsal of the word list.

47 + 36 =	83 - 29 =	6 × 7 =	54 ÷ 9 =	28 + 45 =
54 + 18 =	91 - 47 =	8 × 9 =	72 ÷ 8 =	63 + 19 =
26 + 57 =	72 - 38 =	4 × 12 =	45 ÷ 5 =	37 + 26 =
63 + 29 =	85 - 56 =	7 × 8 =	63 ÷ 7 =	52 + 31 =
38 + 45 =	64 - 27 =	9 × 6 =	48 ÷ 6 =	44 + 29 =
52 + 39 =	93 - 48 =	5 × 11 =	81 ÷ 9 =	67 + 18 =

STEP 2: WORD RECALL (WRITE ALL THE WORDS YOU REMEMBER)

1	2	3	4	5
6	7	8	9	10

Write words in any order — the numbers are just to help you count.

✍ **TOTAL WORDS RECALLED: _____ / 10**

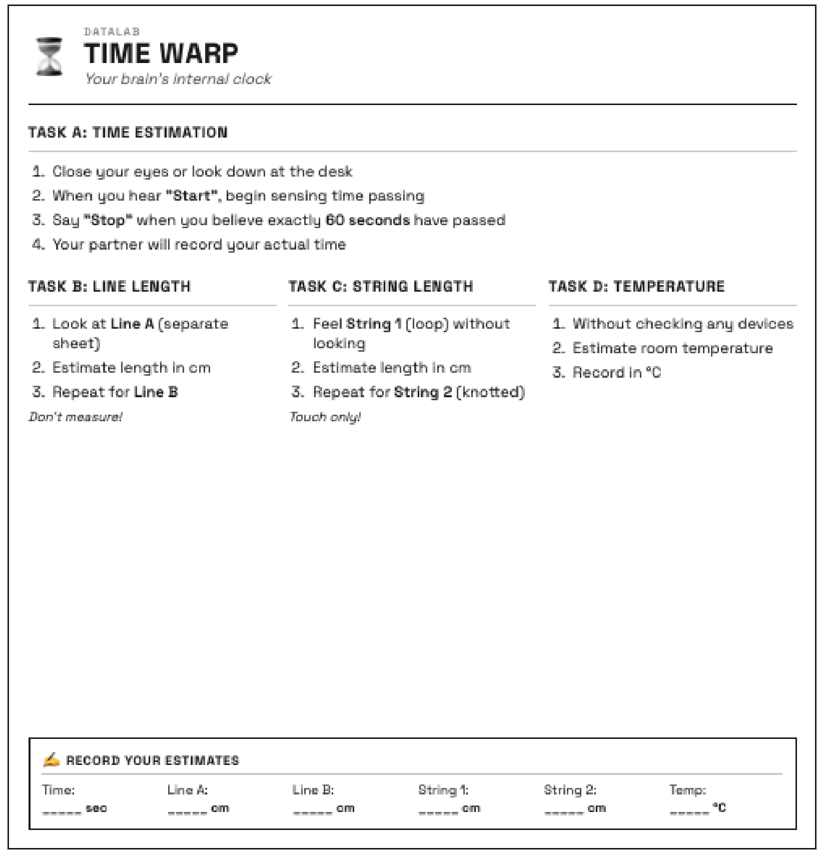
Time Warp

Your brain constructs time

- Most people over- or under-estimate
- Touch vs vision — which is better?

What might have affected your data?

- Did you count in your head?
- When exactly did the stopwatch start?
- Could you see any clocks?



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TIME WARP

Your brain's internal clock

TASK A: TIME ESTIMATION

1. Close your eyes or look down at the desk
2. When you hear **"Start"**, begin sensing time passing
3. Say **"Stop"** when you believe exactly **60 seconds** have passed
4. Your partner will record your actual time

TASK B: LINE LENGTH

1. Look at **Line A** (separate sheet)
2. Estimate length in cm
3. Repeat for **Line B**

Don't measure!

TASK C: STRING LENGTH

1. Feel **String 1** (loop) without looking
2. Estimate length in cm
3. Repeat for **String 2** (knotted)

Touch only!

TASK D: TEMPERATURE

1. Without checking any devices
2. Estimate room temperature
3. Record in °C

✍ **RECORD YOUR ESTIMATES**

Time:	Line A:	Line B:	String 1:	String 2:	Temp:
_____ sec	_____ cm	_____ cm	_____ cm	_____ cm	_____ °C

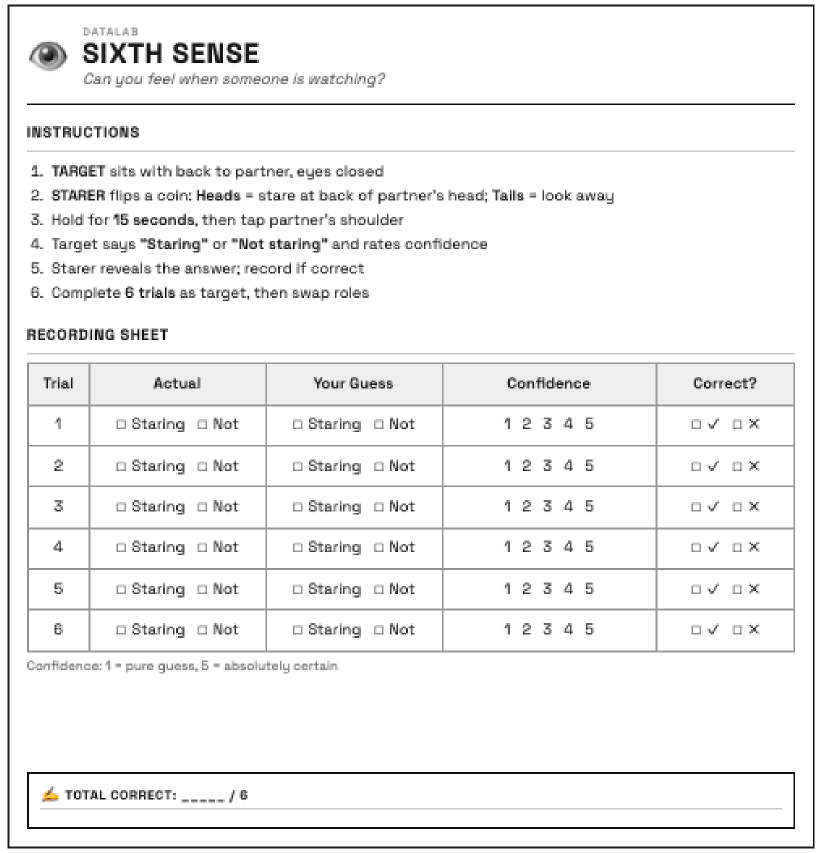
Sixth Sense

Can you detect being watched?

- Chance = 3 out of 6 correct
- Signal detection: hits, misses, false alarms

What might have affected your data?

- Peripheral vision leakage?
- Sound cues from your partner?
- Response bias (saying “yes” more)?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SIXTH SENSE

Can you feel when someone is watching?

INSTRUCTIONS

1. **TARGET** sits with back to partner, eyes closed
2. **STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
3. Hold for **15 seconds,** then tap partner's shoulder
4. Target says **"Staring"** or **"Not staring"** and rates confidence
5. Starer reveals the answer; record if correct
6. Complete **6 trials** as target, then swap roles

RECORDING SHEET

Trial	Actual	Your Guess	Confidence	Correct?
1	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
2	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
3	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
4	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
5	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
6	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕

Confidence: 1 = pure guess, 5 = absolutely certain

✍ TOTAL CORRECT: _____ / 6

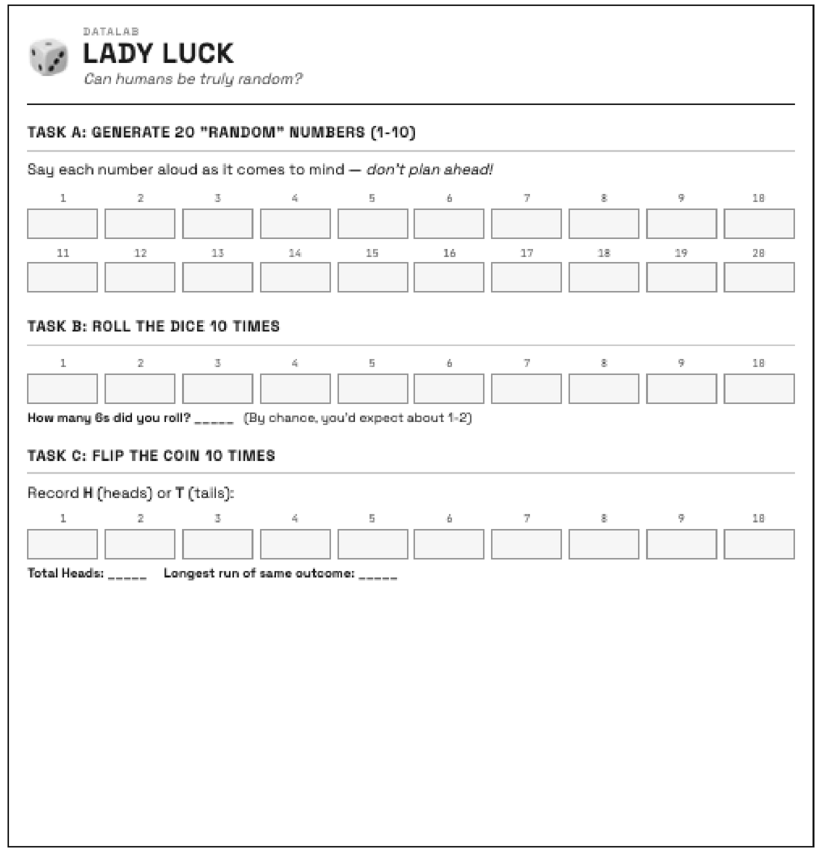
Lady Luck

Humans are bad at random!

- We avoid repeating numbers
- Real randomness has streaks
- Gambler’s fallacy: coins don’t remember

What might have affected your data?

- Number preferences (7 is popular!)
- Did you try to “be random”?
- Were dice/coins recorded accurately?



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LADY LUCK

Can humans be truly random?

TASK A: GENERATE 20 "RANDOM" NUMBERS (1-10)

Say each number aloud as it comes to mind — *don't plan ahead!*

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

TASK B: ROLL THE DICE 10 TIMES

1	2	3	4	5	6	7	8	9	10

How many 6s did you roll? _____ (By chance, you'd expect about 1-2)

TASK C: FLIP THE COIN 10 TIMES

Record **H** (heads) or **T** (tails):

1	2	3	4	5	6	7	8	9	10

Total Heads: _____ **Longest run of same outcome: _____**

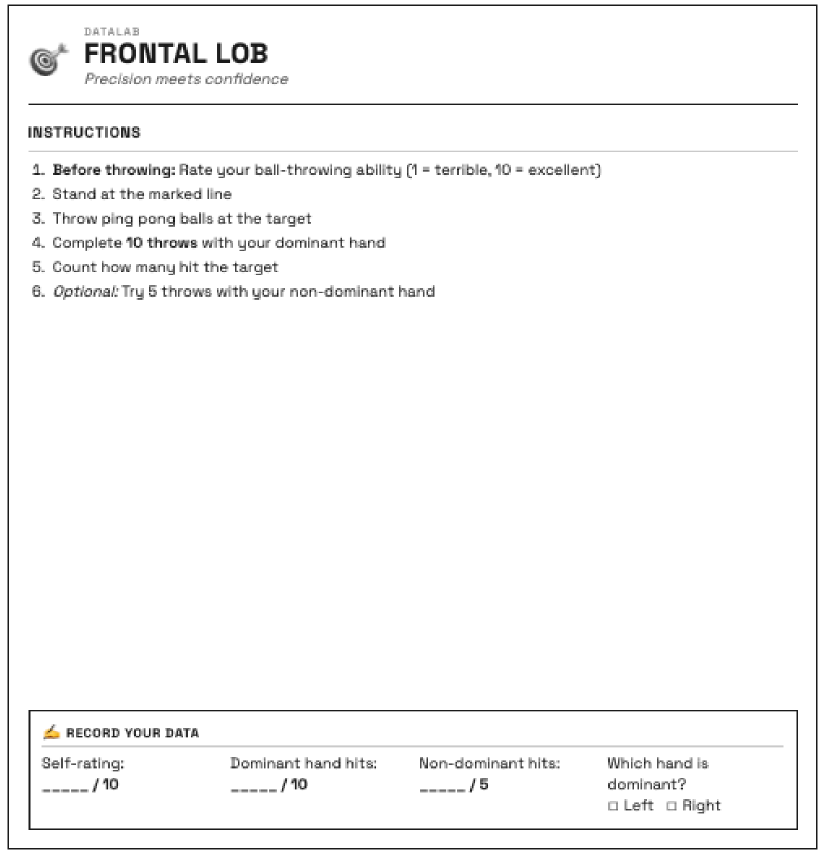
Frontal Lob

How well do you know yourself?

- Calibration = prediction vs reality
- Metacognition: thinking about thinking

What might have affected your data?

- How was “success” defined?
- Self-serving bias in ratings?
- Was throwing distance consistent?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FRONTAL LOB

Precision meets confidence

INSTRUCTIONS

1. **Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
2. Stand at the marked line
3. Throw ping pong balls at the target
4. Complete **10 throws** with your dominant hand
5. Count how many hit the target
6. *Optional:* Try 5 throws with your non-dominant hand

✍ **RECORD YOUR DATA**

Self-rating:	Dominant hand hits:	Non-dominant hits:	Which hand is dominant?
_____ / 10	_____ / 10	_____ / 5	☐ Left ☐ Right

Why This Matters

Collecting data is hard.

These challenges are why researchers:

1. Use **standardised procedures**
2. Define variables **operationally**
3. **Train testers** for consistency
4. **Control the environment**

When we analyse your data, we'll consider these factors.

Data Collected Today

Task	Variables
Memory	Words recalled, positions, intrusions
Reflex	Reaction times (5 trials)
Time Warp	Time, visual, tactile estimates
Sixth Sense	Detection accuracy, confidence
Lady Luck	Number patterns, dice, coins
Frontal Lob	Self-rating, actual performance
Pre/Post	Mood, energy, stress

Summary

What We Did Today

1. Completed surveys (mood, personality)
2. Memory test together
3. 5 tasks as participant and tester
4. Experienced variability firsthand
5. Created pre-post design

Next Week

Week 12: Your First Lecture

Wednesday 10:00–12:00

- Analyse DataLab results together
- Introduction to distributions
- Your data = teaching examples

See you Wednesday!